MME 351 - Marine Electrical Technology

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Course aim

- To provide insight into electrical technology used in marine vessels.







Course Content

- **Power generation and distribution:** Selection of DC and AC generators, Merits and Demerits of AC and DC. Construction & Characteristics of brushless high speed AC generators with automatic Voltage regulator.
- **Rules and Regulations governing electrical machinery on ships**, location and installation of generatos sets. Rating and characteristics of main switch boards. Arrangements of preferential trips and protective devices with wiring diagrams.
- **System earthing:** Isolated neutral systems, resistance earthed system.
- **Motor and Control Equipment:** Characteristics of continuous maximum rated machines. Centralized control of motors in machinery spaces. Classes of insulation (A, B and E) and Temperature rise.

Course Content

- **Motor and Control Equipment:** Special requirements of motors and starters for anchor Windlass & Capstan, wiring diagram, lowering & hoisting arrangement, overload protection, magnetic disc, brakes. Centralised control of motors in machinery spaces. Classes of insulation (A, B and E) and Temperature rise.
- **Starters for miscellaneous electrical equipment** for machinery space auxiliaries eg. Sequential starting and cut outs for an automatic fired boiler incorporating safety devices & combustion control equipment. Sequential starting for refrigerating plants, incorporating safety devices like high pressure cut out, cooling water failure cut out, automatic defrosting with help of timer.

Assessment

- **Continuous assessment** (Assignments, quizzes, mid-sem) - 30%
- **End-of-Semester Examination** (MCQs, Written paper) - 70%

Power Generation and Distribution in Marine Vessels



Uses of electricity in watercraft



Lighting



Pumps

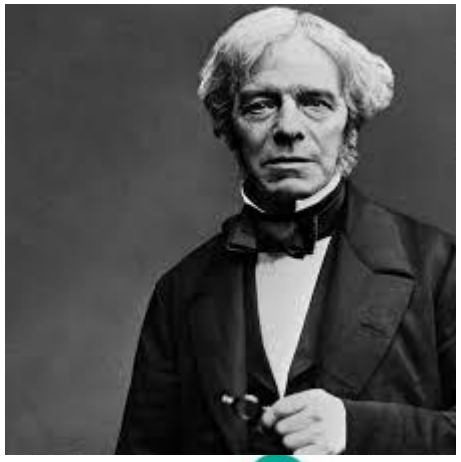


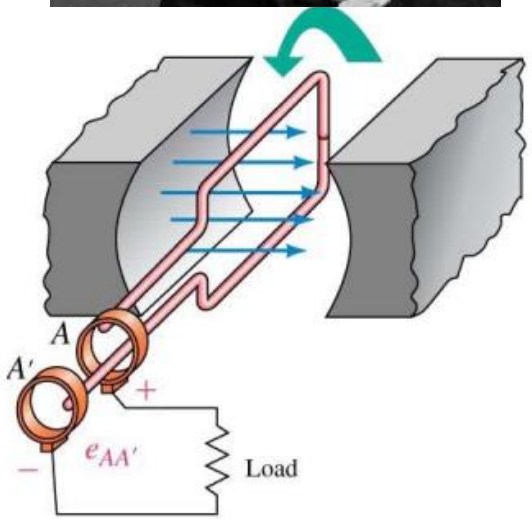


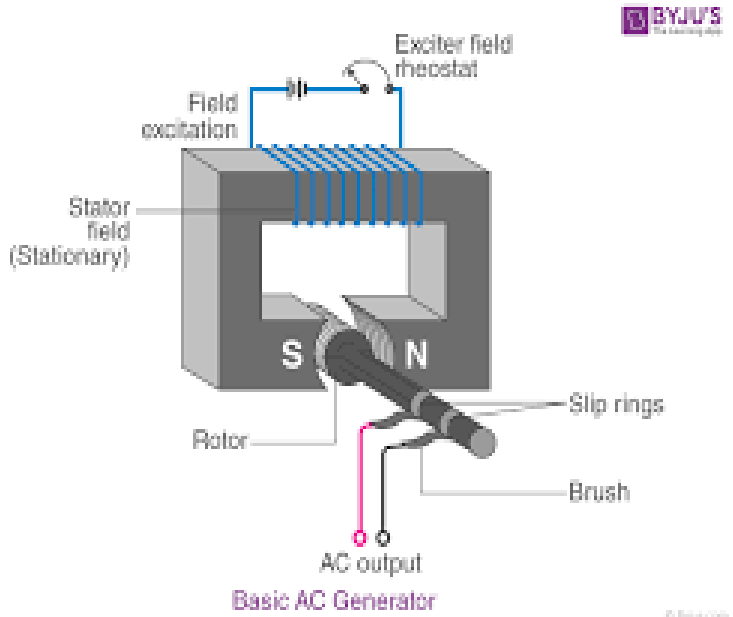
Refrigeration

AC generators

Operating principle of AC generators





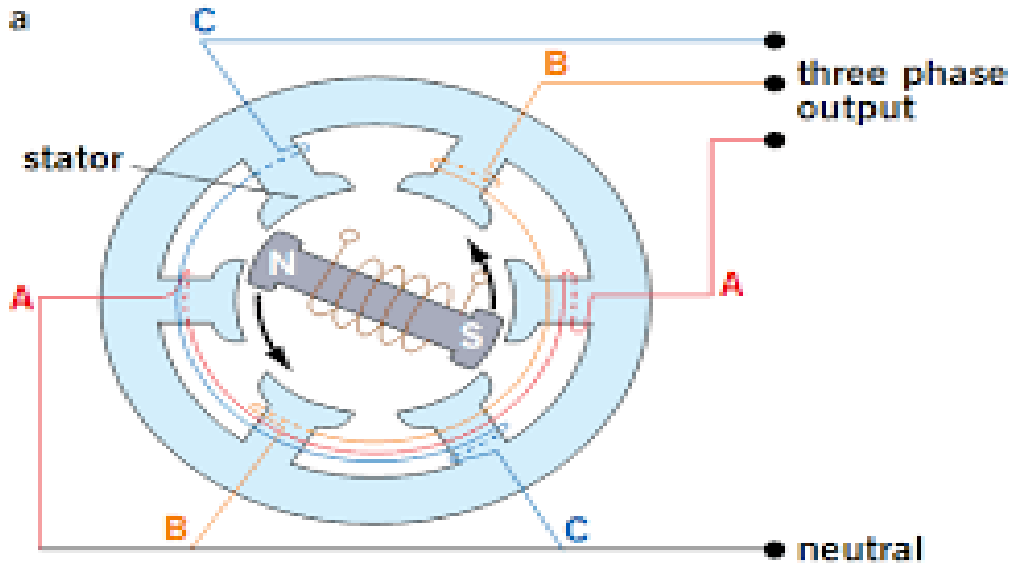




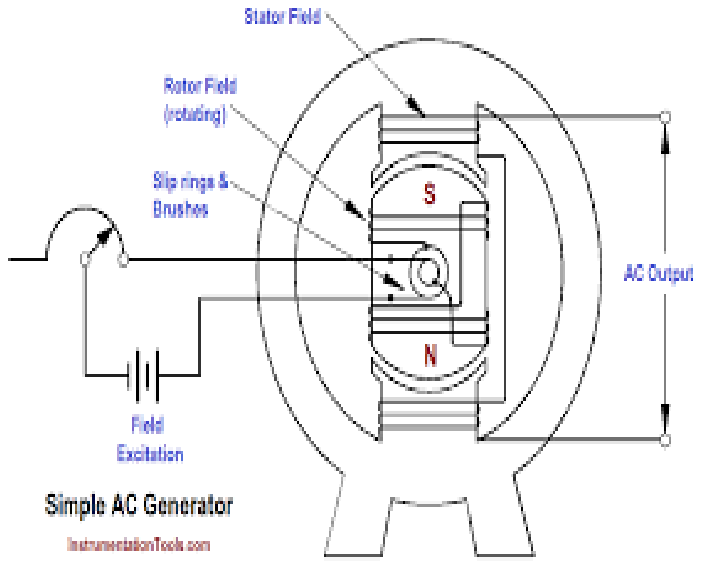


AC generators

Types AC generators



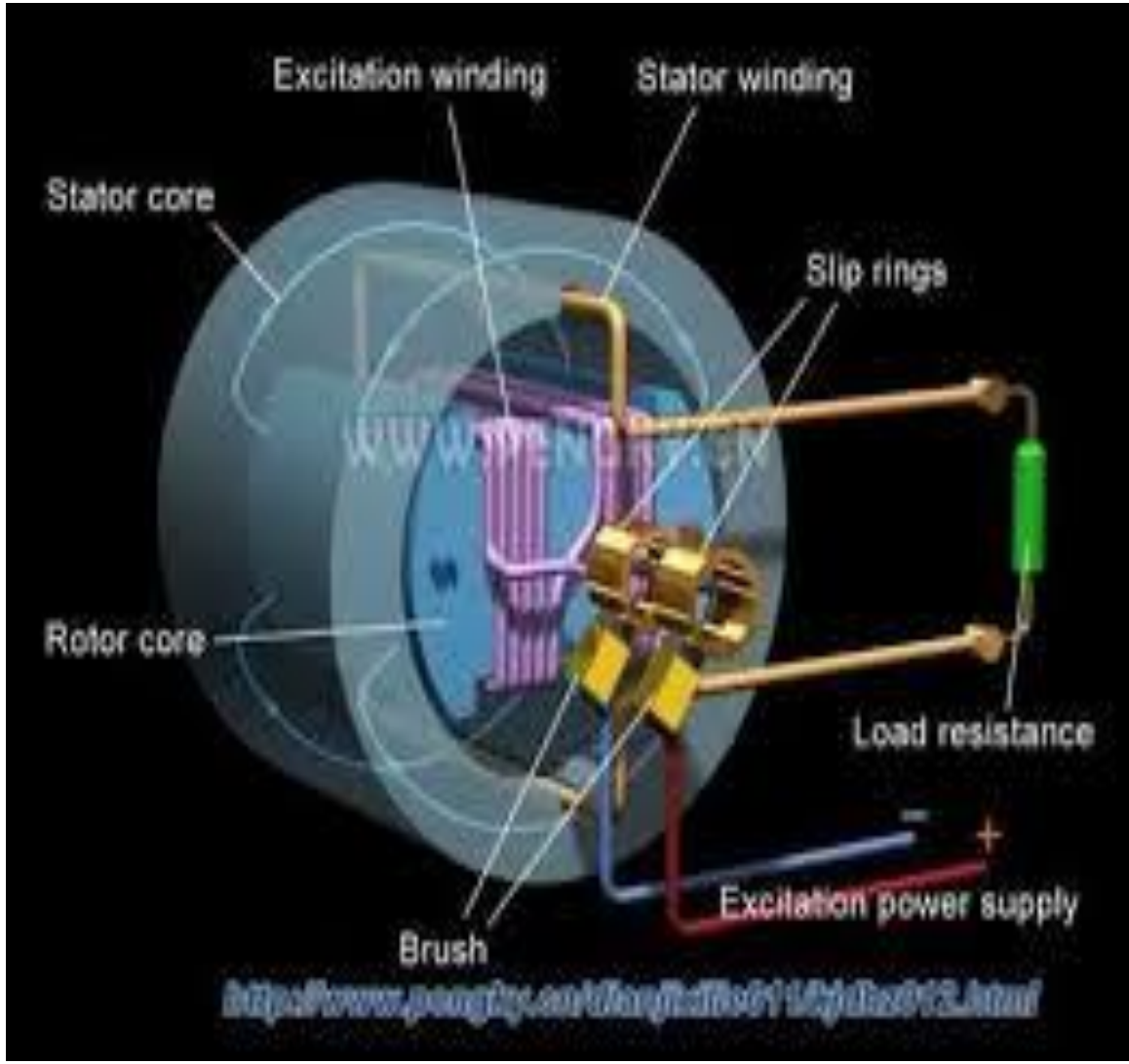
3-phase AC generator



1-phase AC generator

AC generators

- **Major components**
  - **Stator**
  - **Rotor**
  - **Slip rings**

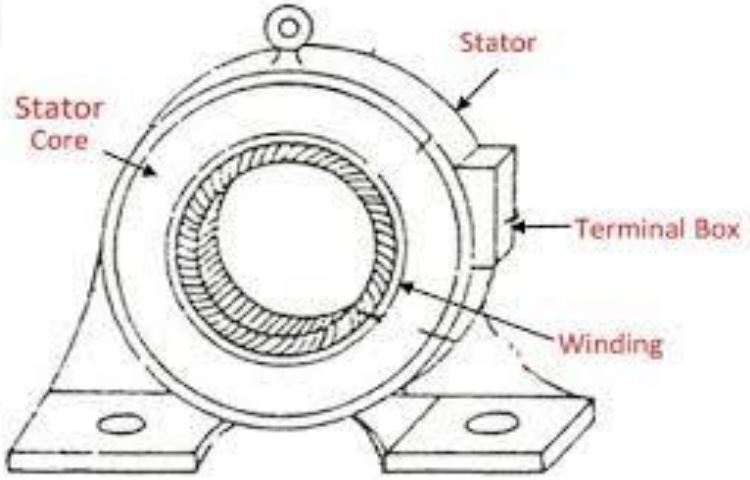


AC generators

- **Stator**











AC generators

- **Rotor**

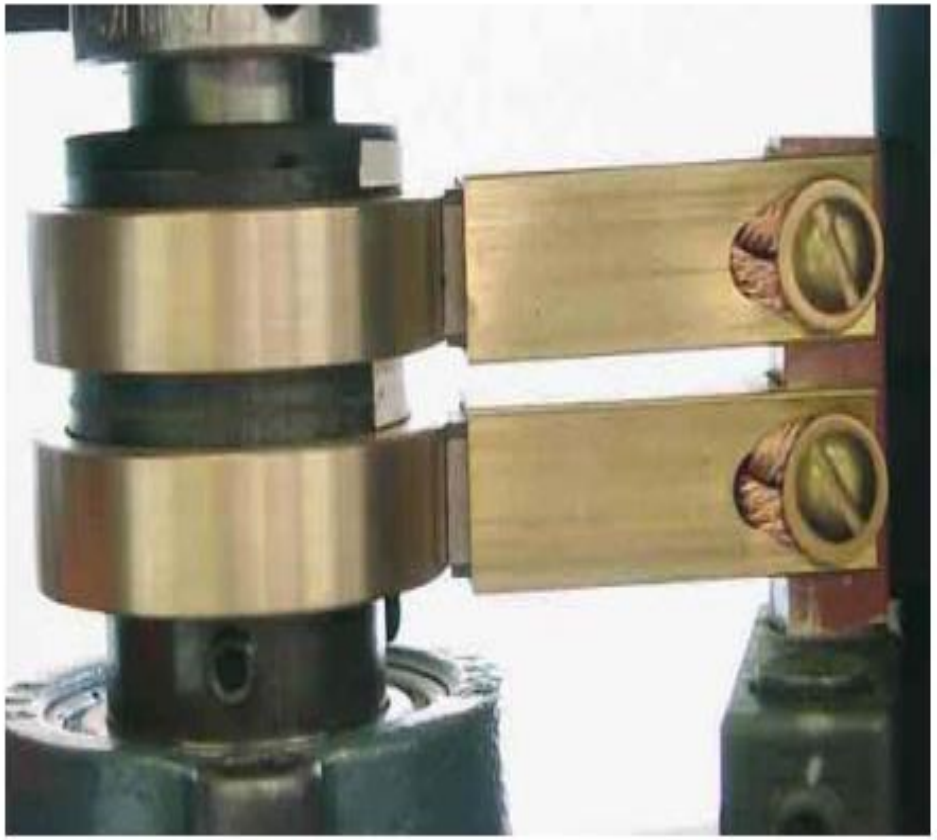






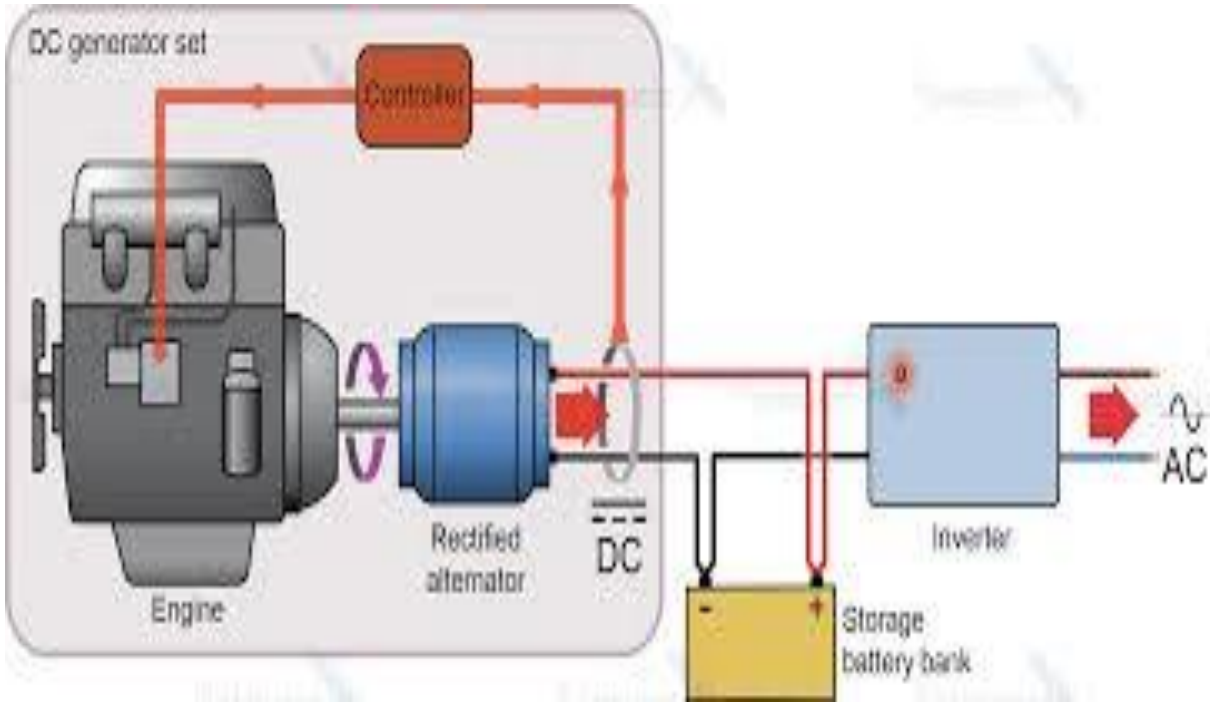
AC generators

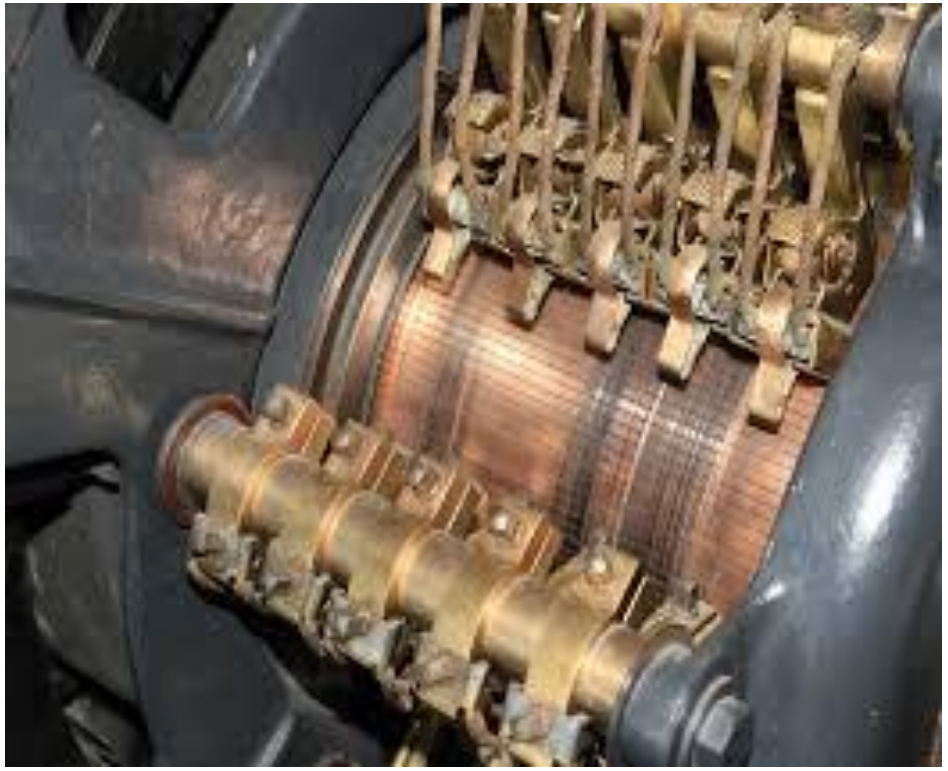
- **Slip rings**



DC generators

- **A DC generator is essentially a 1-phase AC generator with commutator**





AC vrs DC generators

- A DC generator is essentially a 1-phase AC generator with commutator